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Lens assembly

Abstract

A lens assembly includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a reflective element. The first lens is with refractive power and includes a concave surface facing an object side along an axis. The second lens is with refractive power and includes a convex surface facing the object side along the axis. The third, fourth, and fifth lenses are with refractive power. The reflective element includes a reflective surface. The first, second, third, fourth, and fifth lenses are arranged in order from the object side to an image side along the axis. The reflective element is disposed between the first lens and the fifth lens. The lens assembly satisfies: $2\text{ mm} < L < 6\text{ mm}$; wherein L is an interval from an object side surface of the first lens to the reflective surface along the axis.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The invention relates to a lens assembly.

Description of the Related Art

(2) The current development trend of a lens assembly for mobile phone is toward high resolution. The number of lenses used in the lens assembly for mobile phone is increasing, making the total length and outer diameter of the lens assembly for mobile phone is getting longer and larger and the ratio of the volume of the lens assembly to the internal volume of the mobile phone is also getting bigger, which can no longer meet the requirements of thin and light for mobile phone. Therefore, the lens assembly needs a new structure in order to meet the requirements of high resolution and miniaturization at the same time.

BRIEF SUMMARY OF THE INVENTION

(3) The invention provides a lens assembly to solve the above problems. The lens assembly of the invention is provided with characteristics of a shortened total lens length, a decreased outer diameter, an increased resolution, an easy manufacture process, and still has a good optical performance.

(4) The lens assembly in accordance with an exemplary embodiment of the invention includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a reflective element. The first lens is with refractive power and includes a concave surface facing an object side along an axis. The second lens is with refractive power and includes a convex surface facing the object side along the axis. The third lens is with refractive power. The fourth lens is with refractive power. The fifth lens is with refractive power. The reflective element includes a reflective surface. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from the object side to an image side along the axis. The reflective element is disposed between the first lens and the fifth lens. The lens assembly satisfies: $2\text{ mm} < L < 6\text{ mm}$; wherein L is an interval from an object side surface of the first lens to the reflective surface along the axis.

(5) In another exemplary embodiment, the first lens is with negative refractive power, the second lens is with positive refractive power, the third lens is with positive refractive power, the fourth lens is with negative refractive power, and the fifth lens is with positive refractive power.

(6) In yet another exemplary embodiment, the first lens further includes a plane surface or a concave surface facing the image side along the axis, the third lens includes a convex surface facing the object side along the axis, and the fourth lens is a meniscus lens.

(7) In another exemplary embodiment, the fifth lens includes a convex surface facing the image side along the axis and another convex surface or a concave surface facing the object side along the axis.

(8) In yet another exemplary embodiment, the second lens is a biconvex lens and further includes another convex surface facing the image side along the axis, the third lens is a meniscus lens and further includes a concave surface facing the image side along the axis, and the fourth lens includes a concave surface facing the object side along the axis and a convex surface facing the image side along the axis.

(9) In another exemplary embodiment, the second lens is a meniscus lens and further includes a concave surface facing the image side along the axis, the third lens is a biconvex lens and further includes a convex surface facing the image side along the axis, and the fourth lens includes a convex surface facing the object side along the axis and a concave surface facing the image side along the axis.

(10) In yet another exemplary embodiment, the lens assembly satisfies at least any one of the following conditions: $0.1 < SD5/TTL < 0.6$; $3.5 < TTL/SD1 < 14$; $0.5 < SD1/L1T < 3$; wherein SD1 is an effective optical diameter of the first lens, SD5 is an effective optical diameter of the fifth lens, TTL is an interval from the object side surface of the first lens to an image plane along the axis, and L1T is a thickness along the axis of the first lens.

(11) In another exemplary embodiment, the reflective element further includes an incident surface facing the object side along the axis and an exit surface facing the image side along the axis and the lens assembly satisfies at least any one of the following conditions: $0.5 < MT/L1T < 10$;

$0 < MT / (SD2 + SD3 + SD4 + SD5) < 1$; wherein MT is an interval from the incident surface along the axis, L1T is a thickness along the axis of the first lens, SD2 is an effective optical diameter of the second lens, SD3 is an effective optical diameter of the third lens, SD4 is an effective optical diameter of the fourth lens, and SD5 is an effective optical diameter of the fifth lens.

(12) In yet another exemplary embodiment, the lens assembly satisfies at least any one of the following conditions: $0 < f / L1T < 5$; $1 < TTL / L < 5$; $0 < f / L < 2$; $-20 < R.sub.11 / L1T < 0$; wherein TTL is an interval from the object side surface of the first lens to an image plane along the axis, L is the interval from the object side surface of the first lens to the reflective surface along the axis, f is an effective focal length of the lens assembly, R.sub.11 is a radius of curvature of the object side surface of the first lens, and L1T is a thickness along the axis of the first lens.

(13) In another exemplary embodiment, the lens assembly satisfies at least any one of the following conditions: $2 < ALD / f < 8$; $-12 < f.sub.1 / L1T < 0$; wherein ALD is a total sum of the effective optical diameter of all lenses of the lens assembly, f is an effective focal length of the lens assembly, f.sub.1 is an effective focal length of the first lens, and L1T is a thickness along the optical axis of the first lens.

(14) In yet another exemplary embodiment, the first lens is a meniscus lens with positive refractive power, the second lens is with negative refractive power, and the fourth lens is with positive refractive power.

(15) In another exemplary embodiment, the first lens further includes a convex surface facing the image side along the axis, the second lens is a meniscus lens and further includes a concave surface facing the image side along the axis, and the fifth lens is a meniscus lens and includes a convex surface facing the object side along the axis and a concave surface facing the image side along the axis.

(16) In yet another exemplary embodiment, the fourth lens includes a convex surface facing the image side along the axis and another convex surface or a concave surface facing the object side along the axis.

(17) In another exemplary embodiment, the third lens is a biconvex lens with positive refractive power and includes a convex surface facing the object side along the axis and another convex surface facing the image side along the axis and the fifth lens is with positive refractive power.

(18) In yet another exemplary embodiment, the third lens is a meniscus lens with negative refractive power and includes a convex surface facing the object side along the axis and a concave surface facing the image side along the axis and the fifth lens is with positive refractive power.

(19) In another exemplary embodiment, the third lens is a meniscus lens with negative refractive power and includes a concave surface facing the object side along the axis and a convex surface facing the image side along the axis and the fifth lens is with negative refractive power.

(20) In yet another exemplary embodiment, the lens assembly satisfies at least any one of the following conditions: $-20 < R.sub.11 / L1T < 0$; $3.5 < TTL / SD1 < 14$; wherein R.sub.11 is a radius of curvature of the object side surface of the first lens, L1T is a thickness along the axis of the first lens, TTL is an interval from the object side surface of the first lens to an image plane along the axis, and SD1 is an effective optical diameter of the first lens.

(21) In another exemplary embodiment, the lens assembly satisfies at least any one of the following conditions: $3 < (f.sub.1 + f) / SD1 < 5$; $-6 < R.sub.11 / R.sub.52 < -0.5$; $1 < f.sub.1 / L1T < 4$; $1 \text{ mm.sup.2} < f.sub.1 \times f.sub.4 < 11 \text{ mm.sup.2}$; $-12 \text{ mm.sup.2} < f.sub.2 \times f.sub.4 < -1 \text{ mm.sup.2}$; wherein R.sub.11 is a radius of curvature of the object side surface of the first lens, R.sub.52 is a radius of curvature of an image side surface of the fifth lens, f is an effective focal length of the lens assembly, f.sub.1 is an effective focal length of the first lens, f.sub.2 is an effective focal length of the second lens, f.sub.4 is an effective focal length of the fourth lens, L1T is a thickness along the axis of the first lens, and SD1 is an effective optical diameter of the first lens.

(22) In yet another exemplary embodiment, the lens assembly satisfies the following condition: $0.5 < f.sub.1 / L < 2.5$; wherein f.sub.1 is an effective focal length of the first lens and L is the interval from the object side surface of the first lens to the reflective surface along the axis.

(23) A detailed description is given in the following embodiments with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:
- (2) FIG. 1 is a lens layout and optical path diagram of a lens assembly in accordance with a first embodiment of the invention;
- (3) FIG. 2A, FIG. 2B, and FIG. 2C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the first embodiment of the invention;
- (4) FIG. 3 is a lens layout and optical path diagram of a lens assembly in accordance with a second embodiment of the invention;
- (5) FIG. 4A, FIG. 4B, and FIG. 4C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the second embodiment of the invention;
- (6) FIG. 5 is a lens layout and optical path diagram of a lens assembly in accordance with a third embodiment of the invention;
- (7) FIG. 6A, FIG. 6B, and FIG. 6C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the third embodiment of the invention;
- (8) FIG. 7 is a lens layout and optical path diagram of a lens assembly in accordance with a fourth embodiment of the invention;
- (9) FIG. 8A, FIG. 8B, and FIG. 8C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the fourth embodiment of the invention;
- (10) FIG. 9 is a lens layout and optical path diagram of a lens assembly in accordance with a fifth embodiment of the invention;
- (11) FIG. 10A, FIG. 10B, and FIG. 10C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the fifth embodiment of the invention;
- (12) FIG. 11 is a lens layout and optical path diagram of a lens assembly in accordance with a sixth embodiment of the invention;
- (13) FIG. 12A, FIG. 12B, and FIG. 12C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the sixth embodiment of the invention;
- (14) FIG. 13 is a lens layout and optical path diagram of a lens assembly in accordance with a seventh embodiment of the invention;
- (15) FIG. 14A, FIG. 14B, and FIG. 14C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the seventh embodiment of the invention;
- (16) FIG. 15 is a lens layout and optical path diagram of a lens assembly in accordance with an eighth embodiment of the invention; and
- (17) FIG. 16A, FIG. 16B, and FIG. 16C depict a field curvature diagram, a distortion diagram, and a modulation transfer function diagram of the lens assembly respectively in accordance with the eighth embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (18) The following description is made for the purpose of illustrating the general principles of the invention and should not be taken in a limiting sense. The scope of the invention is best determined by reference to the appended claims.
- (19) The present invention provides a lens assembly including a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a reflective element. The first lens is with refractive power and includes a concave surface facing an object side along an axis. The second lens is with refractive power and includes a convex surface facing the object side along the axis. The third lens is with refractive power. The fourth lens is with refractive power. The fifth lens is with refractive power. The reflective element includes a reflective surface. The first lens, the second lens, the third lens, the fourth lens, and the fifth lens are arranged in order from the object side to the image side along the axis. The reflective element is disposed between the first lens and the fifth lens. The lens assembly satisfies: $2 \text{ mm} < L < 6 \text{ mm}$; wherein L is an interval from an object side surface of the first lens to

the reflective surface along the axis.

(20) Referring to Table 1, Table 2, Table 4, Table 5, Table 7, Table 8, Table 10, and Table 11, wherein Table 1, Table 4, Table 7, and Table 10 show optical specification in accordance with a first, second, third, and fourth embodiments of the invention respectively and Table 2, Table 5, Table 8, and Table 11 show aspheric coefficients of each aspheric lens in Table 1, Table 4, Table 7, and Table 10 respectively. FIG. 1, FIG. 3, FIG. 5, and FIG. 7 are lens layout and optical path diagrams of the lens assemblies in accordance with the first, second, third, and fourth embodiments of the invention respectively.

(21) The first lenses L11, L21, L31, L41 are with negative refractive power and made of glass material, wherein the object side surfaces S14, S24, S34, S44 are concave surfaces.

(22) The reflective elements P1, P2, P3, P4 are made of glass or plastic material, wherein the incident surfaces S16, S26, S36, S46 are plane surfaces, the reflective surfaces S17, S27, S37, S47 are plane surfaces, and the exit surfaces S18, S28, S38, S48 are plane surfaces. The reflective element can also be a prism or a reflective mirror. The reflective element can also only include one reflective surface when the reflective element is a reflective mirror.

(23) The second lenses L12, L22, L32, L42 are with positive refractive power and made of plastic material, wherein the object side surfaces S19, S29, S39, S49 are convex surfaces and both of the object side surfaces S19, S29, S39, S49 and image side surfaces S110, S210, S310, S410 are aspheric surfaces.

(24) The third lenses L13, L23, L33, L43 are with positive refractive power and made of plastic material, wherein the object side surfaces S111, S211, S311, S411 are convex surfaces and both of the object side surfaces S111, S211, S311, S411 and image side surfaces S112, S212, S312, S412 are aspheric surfaces.

(25) The fourth lenses L14, L24, L34, L44 are with negative refractive power and made of plastic material, wherein both of the object side surfaces S113, S213, S313, S413 and image side surfaces S114, S214, S314, S414 are aspheric surfaces.

(26) The fifth lenses L15, L25, L35, L45 are with positive refractive power and made of plastic material, wherein the image side surfaces S116, S216, S316, S416 are convex surfaces and both of the object side surfaces S115, S215, S315, S415 and image side surfaces S116, S216, S316, S416 are aspheric surfaces.

(27) In addition, the lens assemblies 1, 2, 3, 4 satisfy at least any one of the following conditions:

$$2\text{ mm} < L < 6\text{ mm}; \quad (1)$$

$$3.5 < \text{TTL}/\text{SD1} < 14; \quad (2)$$

$$0.5 < \text{MT}/\text{L1T} < 10; \quad (3)$$

$$0 < \text{MT}/(\text{SD2} + \text{SD3} + \text{SD4} + \text{SD5}) < 1; \quad (4)$$

$$-20 < \text{R.sub.11}/\text{L1T} < 0; \quad (5)$$

$$-12 < \text{f.sub.1}/\text{L1T} < 0; \quad (6)$$

$$0.1 < \text{SD5}/\text{TTL} < 0.6; \quad (7)$$

$$1 < \text{TTL}/L < 5; \quad (8)$$

$$0 < \text{f7L} < 2; \quad (9)$$

$$0.5 < \text{SD1}/\text{L1T} < 3; \quad (10)$$

$$2 < \text{ALD}/\text{f} < 8; \quad (11)$$

$$0 < \text{f}/\text{L1T} < 5; \quad (12)$$

(28) wherein L is an interval from the object side surfaces S14, S24, S34, S44 of the first lenses L11, L21, L31, L41 to the reflective surfaces S17, S27, S37, S47 along the axes OA1, OA2, OA3, OA4 respectively for the first to fourth embodiments, MT is an interval from the incident surfaces S16, S26, S36, S46 to the exit surfaces S18, S28, S38, S48 along the axes OA1, OA2, OA3, OA4 respectively for the first to fourth embodiments, SD1 is an effective optical diameter of the first lenses L11, L21, L31, L41 for the first to fourth embodiments, SD2 is an effective optical diameter of the second lenses L12, L22, L32, L42 for the first to fourth embodiments, SD3 is an effective optical diameter of the third lenses L13, L23, L33, L43 for the first to fourth embodiments, SD4 is an effective optical diameter of the fourth lenses L14, L24, L34, L44 for the first to fourth embodiments, SD5 is an effective optical diameter of the fifth lenses L15, L25, L35, L45 for the first to fourth embodiments, f is an effective focal length of the lens assemblies 1, 2, 3, 4 for the first to fourth embodiments, f.sub.1 is an effective focal length of the first lenses L11, L21, L31, L41 for the first to fourth embodiments, L1T is a thickness along the axes OA1, OA2, OA3, OA4 of the first lenses L11, L21, L31, L41 for the first to fourth embodiments, that is, an interval from the object side surfaces S14, S24, S34, S44 of the first lenses L11, L21, L31, L41 to the image side surfaces S15, S25, S35, S45 along the axes OA1, OA2, OA3, OA4, TTL is an interval from the object side surfaces S14, S24, S34, S44 of the first lenses L11, L21, L31, L41 to the image planes IMA1, IMA2, IMA3, IMA4 along the axes OA1, OA2, OA3, OA4 respectively for the first to fourth embodiments, R.sub.11 is a radius of curvature of the object side surfaces S14, S24, S34, S44 of the first lenses L11, L21, L31, L41 for the first to fourth embodiments, and ALD is a total sum of the effective optical diameter of all lenses for the first to fourth embodiments. With the lens assemblies 1, 2, 3, 4 satisfying at least any one of the above conditions (1)-(12), the total lens length can be effectively shortened, the outer diameter can be effectively decreased, the resolution can be effectively increased, the aberration can be effectively corrected, the chromatic aberration can be effectively corrected, and easy to manufacture process.

(29) A detailed description of a lens assembly in accordance with a first embodiment of the invention is as follows. Referring to FIG. 1, the lens assembly 1 includes a cover glass CG1, a stop ST1, a first lens L11, a reflective element P1, a second lens L12, a third lens L13, a fourth lens L14, a fifth lens L15, and an optical filter OF1, all of which are arranged in order from an object side to an image side along an axis OA1. The reflective element P1 includes an incident surface S16, a reflective surface S17, and an exit surface S18, wherein the incident surface S16 and the exit surface S18 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S17 to change the propagation direction and imaged on an image plane IMA1. The image plane IMA1 and the exit surface S18 are parallel to each other. In the first embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(30) According to the foregoing, wherein: both of the object side surface S11 and image side surface S12 of the cover glass CG1 are plane surfaces; the first lens L11 is a plane-concave lens, wherein the image side surface S15 is a plane surface and the object side surface S14 is a spherical surface; the second lens L12 is a biconvex lens, wherein the image side surface S110 is a convex surface; the third lens L13 is a meniscus lens, wherein the image side surface S112 is a concave surface; the fourth lens L14 is a meniscus lens, wherein the object side surface S113 is a concave surface and the image side surface S114 is a convex surface; the fifth lens L15 is a biconvex lens, wherein the object side surface S115 is a convex surface; and both of the object side surface S117 and image side surface S118 of the optical filter OF1 are plane surfaces.

(31) With the above design of the lenses, stop ST1, reflective element P1, and at least any one of the conditions (1)-(12) satisfied, the lens assembly 1 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(32) Table 1 shows the optical specification of the lens assembly 1 in FIG. 1.

(33) TABLE-US-00001 TABLE 1 Effective Focal Length = 2.36146 mm F-number = 2.4 Total Lens Length = 10.52 mm Field of View = 78 degrees Radius of Effective Surface Curvature Thickness Focal Length Number (mm) (mm) Nd Vd (mm) Remark S11 ∞ 0.75 1.5168 64.167336 CG1 S12 ∞ 0.2 S13 ∞ 0.05010699 ST1 S14 -6.399133 1 2.003303 28.319976 -6.33767 L11 S15 ∞ 0 S16 ∞ 1.5 2.003303 28.319976 P1 Incident Surface S17 ∞ 1.5 2.003303 28.319976 P1 Reflective Surface S18 ∞ 0.05 P1 Exit Surface S19 4.30236 1.220212 1.535218 56.115254 3.89889 L12 S110

-3.673788 0.025 S111 1.862688 1.127119 1.535218 56.115252 5.012344 L13 S112 4.770468 0.7807939 S113 -0.437913 0.487242 1.671339 19.242889 -1.40602 L14 S114 -1.174612 0.015 S115 0.8410429 0.8553014 1.535081 55.779665 1.54751 L15 S116 -43.79546 0.5 S117 ∞ 0.21 1.5168 64.167336 OF1 S118 ∞ 0.25

(34) The aspheric surface sag z of each aspheric lens in table 1 can be calculated by the following formula:

$$z = ch \cdot \sup.2 / \{1 + [1 -$$

$$(k+1)c \cdot \sup.2h \cdot \sup.2\} \cdot \sup.1/2 + Ah \cdot \sup.4 + Bh \cdot \sup.6 + Ch \cdot \sup.8 + Dh \cdot \sup.10 + Eh \cdot \sup.12 + Fh \cdot \sup.14 + Gh \cdot \sup.16 + Hh \cdot \sup.18 + Ih \cdot \sup.20 + Jh \cdot \sup.3 + Kh \cdot \sup.5 + Lh \cdot \sup.7$$

where c is curvature, h is the vertical distance from the lens surface to the optical axis, k is conic constant and $A, B, C, D, E, F, G, H, I, J, K$ and L are aspheric coefficients.

(35) In the first embodiment, the conic constant k and the aspheric coefficients $A, B, C, D, E, F, G, H, I, J, K, L$ of each aspheric lens are shown in Table 2.

(36) TABLE-US-00002 TABLE 2 Surface A B C D E F Number k G H I J K L S19 -36.04416 -6.4247E-02 5.6985E-02 -2.5117E-02 5.7448E-03 -2.3892E-04 -1.1273E-04 -1.4731E-05 1.1584E-05 -1.0947E-06 -1.4817E-02 -1.4010E-03 -1.0969E-03 S110 2.6602 1.2024E-01 -4.5800E-02 1.1232E-02 -1.7176E-03 -7.2719E-05 4.5098E-05 1.0977E-05 -2.2770E-06 -1.5996E-07 -1.1107E-01 2.4481E-02 -6.0939E-03 S111 -0.3378937 1.3195E-01 -6.0139E-02 2.4629E-02 -5.2884E-03 3.1803E-04 -1.3230E-04 3.0679E-05 2.0153E-05 -4.6069E-06 -9.0329E-02 2.9799E-02 -1.6650E-03 S112 1.034241 -3.7632E-02 1.8802E-02 -1.6136E-02 5.0841E-03 -6.1590E-04 -1.6311E-04 7.0473E-06 1.6313E-05 -2.7390E-06 3.1593E-02 8.9310E-03 1.1042E-02 S113 -2.784855 5.1219E-02 -6.1254E-02 2.4460E-02 -1.2739E-03 -1.2216E-03 3.5027E-04 -2.3110E-05 -1.3055E-05 2.1766E-06 -7.7116E-02 -8.7436E-03 6.8449E-04 S114 -4.268764 9.7529E-02 -1.0727E-01 3.0119E-02 -8.9519E-04 -3.0243E-03 9.8545E-04 9.6734E-06 -2.0888E-05 -2.1280E-06 -1.5871E-01 6.7286E-02 -1.0446E-02 S115 -5.17819 2.1008E-01 -9.0676E-02 6.7556E-02 -3.7242E-02 6.3652E-03 -1.0431E-04 1.0062E-03 -5.4799E-04 8.0685E-05 -1.2006E-01 -3.2558E-02 1.1295E-02 S116 652.8874 -3.1732E-02 1.8180E-01 -7.5323E-02 1.0583E-02 -1.1999E-04 -1.3128E-04 -3.1456E-05 9.6217E-05 -2.2518E-05 1.0825E-01 -1.7720E-01 2.1218E-02

(37) Table 3 shows the parameters and condition values for conditions (1)-(12) in accordance with the first embodiment of the invention. It can be seen from Table 3 that the lens assembly 1 of the first embodiment satisfies the conditions (1)-(12).

(38) TABLE-US-00003 TABLE 3 SD1 1.43 mm SD2 3.56 mm SD3 3.63 mm SD4 3.56 mm SD5 3.27 mm MT 3 mm L 2.5 mm ALD 15.45 mm f.sub.1/L1T -6.33767 SD5/TTL 0.310837 TTL/SD1 7.356643 MT/L1T 3 MT/(SD2 + 0.21398 TTL/L 4.208 SD3 + SD4 + SD5) f/L 0.944584 R.sub.11/L1T -6.39913 SD1/L1T 1.43 ALD/f 6.542563 f/L1T 2.36146

(39) In addition, the lens assembly 1 of the first embodiment can meet the requirements of optical performance as seen in FIGS. 2A-2C. It can be seen from FIG. 2A that the field curvature of tangential direction and sagittal direction in the lens assembly 1 of the first embodiment ranges from 0.06 mm to 0.16 mm. It can be seen from FIG. 2B that the distortion in the lens assembly 1 of the first embodiment ranges from -0.5% to 2.5%. It can be seen from FIG. 2C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 1 of the first embodiment ranges from 0.17 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 1 of the first embodiment can be corrected effectively, and the resolution of the lens assembly 1 of the first embodiment can meet the requirement. Therefore, the lens assembly 1 of the first embodiment is capable of good optical performance.

(40) Referring to FIG. 3, the lens assembly 2 includes a cover glass CG2, a stop ST2, a first lens L21, a reflective element P2, a second lens L22, a third lens L23, a fourth lens L24, a fifth lens L25, and an optical filter OF2, all of which are arranged in order from an object side to an image side along an axis OA2. The reflective element P2 includes an incident surface S26, a reflective surface S27, and an exit surface S28, wherein the incident surface S26 and the exit surface S28 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S27 to change the propagation direction and imaged on an image plane IMA2. The image plane IMA2 and the exit surface S28 are parallel to each other. In the second embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(41) According to the foregoing, wherein: both of the object side surface S21 and image side surface S22 of the cover glass CG2 are plane surfaces; the first lens L21 is a plane-concave lens, wherein the image side surface S25 is a plane surface and the object side surface S24 is an aspheric surface; the second lens L22 is a biconvex lens, wherein the image side surface S210 is a convex surface; the third lens L23 is a meniscus lens, wherein the image side surface S212 is a concave surface; the fourth lens L24 is a meniscus lens, wherein the object side surface S213 is a concave surface and the image side surface S214 is a convex surface; the fifth lens L25 is a biconvex lens, wherein the object side surface S215 is a convex surface; and both of the object side surface S217 and image side surface S218 of the optical filter OF2 are plane surfaces.

(42) With the above design of the lenses, stop ST2, reflective element P2, and at least any one of the conditions (1)-(12) satisfied, the lens assembly 2 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(43) Table 4 shows the optical specification of the lens assembly 2 in FIG. 3.

(44) TABLE-US-00004 TABLE 4 Effective Focal Length = 2.55359 mm F-number = 2.4 Total Lens Length = 9.98 mm Field of View = 76 degrees Radius of Effective Surface Curvature Thickness Focal Length Number (mm) (mm) Nd Vd (mm) Remark S21 ∞ 0.75 1.5168 64.167336 CG2 S22 ∞ 0.2 S23 ∞ 0.06 ST2 S24 -7.518558 0.84 2.0033 28.32 -7.44636 L21 S25 0 (mm) S26 1.55 2.0033 28.32 P2 Incident Surface S27 ∞ 1.55 2.0033 28.32 P2 Reflective Surface S28 ∞ 0.015 P2 Exit Surface S29 3.427519 1.227709 1.5352 56.11 5.36455 L22 S210 -15.78844 0.015 S211 1.554132 1.01393 1.5352 56.11 3.05754 L23 S212 22.49015 0.9438974 S213 -0.4246024 0.363 1.670993 19.243508 -1.42553 L24 S214 -1.019583 0.015 S215 1.214293 0.6850868 1.535056 56.110001 1.66977 L25 S216 -2.748457 0.54 S217 ∞ 0.21 1.5168 64.167336 OF2 S218 ∞ 0

(45) The definition of aspheric surface sag z of each aspheric lens in table 4 is the same as that of in Table 1, and is not described here again.

(46) In the second embodiment, the conic constant k and the aspheric coefficients $A, B, C, D, E, F, G, H, I, J, K, L$ of each aspheric lens are shown in Table 5.

(47) TABLE-US-00005 TABLE 5 Surface A B C D E F Number k G H I J K L S24 47.67621 4.3407E-02 -1.0693E+00 1.2987E+01 -7.4448E+01 1.9156E+02 -1.2784E+02 -1.5929E+02 0 0 0 0 0 S29 0.6329401 -2.1170E-02 1.6332E-02 -5.6300E-03 9.6280E-04 2.0490E-04 -1.4287E-04 2.9490E-05 0 0 0 0 0 S210 -100.8146 2.9445E-02 -5.6494E-05 3.3710E-03 -1.1380E-03 -7.1591E-04 3.8440E-04 -4.2654E-05 0 0 0 0 0 S211 -0.145394 3.5712E-02 -1.0172E-02 4.5920E-03 1.9470E-03 -7.1696E-04 -7.6093E-05 1.5400E-04 0 0 0 0 0 S212 196.5748 -3.5260E-02 -2.3773E-02 1.1229E-02 3.1610E-03 -4.6242E-04 -3.4735E-04 5.6650E-05 0 0 0 0 0 S213 -2.099871 -1.1805E-01 3.8600E-02 -1.0134E-02 3.1070E-04 1.1090E-03 -5.8650E-04 9.9210E-05 0 0 0 0 0 S214 -7.217332 4.1566E-02 -8.8390E-02 4.2417E-02 -1.6768E-02 6.3890E-03 -1.7846E-03 2.3700E-04 0 0 0 0 0 S215 -1.000027 2.3287E-01 -1.0349E-01 7.3124E-02 -1.2419E-03 -5.3408E-03 -1.4568E-03 5.1690E-04 0 0 0 0 0 S216 -39.33204 -1.5093E-01 1.4420E-01 -3.3963E-02 5.4170E-03 -1.0227E-03 -9.5227E-04 2.7110E-04 0 0 0 0 0

(48) Table 6 shows the parameters and condition values for conditions (1)-(12) in accordance with the second embodiment of the invention. It can be seen from Table 6 that the lens assembly 2 of the second embodiment satisfies the conditions (1)-(12).

(49) TABLE-US-00006 TABLE 6 SD1 1.09 mm SD2 3.42 mm SD3 3.23 mm SD4 3.08 mm SD5 3.11 mm MT 3.1 mm L 2.39 mm ALD 13.93 mm f.sub.1/L1T -8.86471 SD5/TTL 0.311623 TTL/SD1 9.155963 MT/L1T 3.690476 MT/(SD2 + 0.241433 TTL/L 4.175732 SD3 + SD4 + SD5) f/L 1.068448 R.sub.11/L1T -8.95066 SD1/L1T 1.297619 ALD/f 5.455065 f/L1T 3.039988

(50) In addition, the lens assembly 2 of the second embodiment can meet the requirements of optical performance as seen in FIGS. 4A-4C. It can be seen from FIG. 4A that the field curvature of tangential direction and sagittal direction in the lens assembly 2 of the second embodiment ranges from

−0.4 mm to 0.1 mm. It can be seen from FIG. 4B that the distortion in the lens assembly 2 of the second embodiment ranges from 0% to 3%. It can be seen from FIG. 4C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 2 of the second embodiment ranges from 0.27 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 2 of the second embodiment can be corrected effectively, and the resolution of the lens assembly 2 of the second embodiment can meet the requirement. Therefore, the lens assembly 2 of the second embodiment is capable of good optical performance.

(51) Referring to FIG. 5, the lens assembly 3 includes a cover glass CG3, a stop ST3, a first lens L31, a reflective element P3, a second lens L32, a third lens L33, a fourth lens L34, a fifth lens L35, and an optical filter OF3, all of which are arranged in order from an object side to an image side along an axis OA3. The reflective element P3 includes an incident surface S36, a reflective surface S37, and an exit surface S38, wherein the incident surface S36 and the exit surface S38 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S37 to change the propagation direction and imaged on an image plane IMA3. The image plane IMA3 and the exit surface S38 are parallel to each other. In the third embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(52) According to the foregoing, wherein: both of the object side surface S31 and image side surface S32 of the cover glass CG3 are plane surfaces; the first lens L31 is a plane-concave lens, wherein the image side surface S35 is a plane surface and the object side surface S34 is an aspheric surface; the second lens L32 is a meniscus lens, wherein the image side surface S310 is a concave surface; the third lens L33 is a biconvex lens, wherein the image side surface S312 is a convex surface; the fourth lens L34 is a meniscus lens, wherein the object side surface S313 is a convex surface and the image side surface S314 is a concave surface; the fifth lens L35 is a biconvex lens, wherein the object side surface S315 is a convex surface; and both of the object side surface S317 and image side surface S318 of the optical filter OF3 are plane surfaces.

(53) With the above design of the lenses, stop ST3, reflective element P3, and at least any one of the conditions (1)-(12) satisfied, the lens assembly 3 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(54) Table 7 shows the optical specification of the lens assembly 3 in FIG. 5.

(55) TABLE-US-00007 TABLE 7 Effective Focal Length = 3 mm F-number = 2.8 Total Lens Length = 10.37 mm Field of View = 65 degrees Radius of Effective Surface Curvature Thickness Focal Length Number (mm) (mm) Nd Vd (mm) Remark S31 ∞ 0.75 1.5168 64.167336 CG3 S32 ∞ 0.2 S33 ∞ 0.004 ST3 S34 −11.21434 1 2.003303 28.319976 −11.1065 L31 S35 ∞ 0 S36 ∞ 1.5 2.003303 28.319976 P3 Incident Surface S37 ∞ 1.5 2.003303 28.319976 P3 Reflective Surface S38 ∞ 0.0433509 P3 Exit Surface S39 2.518995 0.3999828 1.670993 19.243508 17.9461 L32 S310 2.973444 0.0419884 S311 6.997376 0.9851131 1.5352 56.11 3.18661 L33 S312 −2.153499 0.0419915 S313 1.037247 0.4499987 1.661342 20.372904 −3.38567 L34 S314 0.5874147 0.4971176 S315 15.72896 0.80001 1.5352 56.11 3.16916 L35 S316 −1.87422 0.9428598 S317 ∞ 0.21 1.5168 64.167336 OF3 S318 ∞ 0.9627068

(56) The definition of aspheric surface sag z of each aspheric lens in table 7 is the same as that of in Table 1, and is not described here again.

(57) In the third embodiment, the conic constant k and the aspheric coefficients A, B, C, D, E, F, G, H, I, J, K, L of each aspheric lens are shown in Table 8.

(58) TABLE-US-00008 TABLE 8 Surface A B C D E F Number k G H I J K L S34 −138.6666 −1.9840E−02 1.9963E−01 −1.4553E+00 3.7350E+00 2.6108E+00 −2.5893E+01 2.9984E+01 0 0 0 0 0 S39 −12.87667 1.7964E−02 −1.1823E−01 8.6661E−02 1.4891E−02 −3.4462E−02 −9.0718E−04 1.0998E−02 −4.3436E−03 7.2260E−04 1.4711E−02 −4.1432E−02 1.7803E−02 S310 −28.27857 −6.8919E−02 3.3268E−02 2.7418E−02 1.4071E−02 −9.0591E−03 −1.0561E−02 −3.9596E−03 8.2680E−03 −2.0196E−03 −1.0718E−02 −1.2483E−01 4.3886E−02 S311 22.26557 −2.3175E−01 9.5566E−02 −2.9471E−04 −6.2191E−02 2.9120E−03 1.7516E−02 5.7870E−03 −1.1111E−02 2.9140E−03 −7.4692E−02 1.4421E−02 7.9804E−02 S312 0.8729461 −3.6232E−01 6.0449E−02 −1.5904E−02 1.4409E−02 4.9020E−03 −3.2073E−03 −2.8463E−03 1.5530E−03 −1.9476E−04 −7.0472E−02 1.5231E−01 −1.9737E−02 S313 −2.594954 1.2251E−01 −2.3435E−02 2.2612E−02 −1.5037E−02 −7.0001E−03 8.1720E−03 4.9770E−03 −5.7978E−03 1.2750E−03 2.3100E−03 −1.0014E−01 3.0803E−02 S314 −3.060501 −3.6878E−01 1.8270E−01 −1.2540E−01 1.1575E−02 4.4544E−02 −5.0402E−03 −1.8848E−02 9.6580E−03 −1.3917E−03 9.5603E−02 2.3883E−01 −7.4220E−02 S315 112.0945 −1.4760E−01 1.0645E−01 8.1370E−03 −4.8291E−02 −7.8726E−03 1.8660E−03 1.6575E−02 −1.0499E−02 1.9420E−03 5.0025E−02 −1.7883E−01 1.0793E−01 S316 0 −3.4445E−01 −3.6371E−01 1.7706E−01 −6.1844E−02 9.7980E−03 −6.0883E−03 6.4610E−03 −2.2497E−03 2.3980E−04 7.2743E−02 3.7552E−01 −3.1586E−03

(59) Table 9 shows the parameters and condition values for conditions (1)-(12) in accordance with the third embodiment of the invention. It can be seen from Table 9 that the lens assembly 3 of the third embodiment satisfies the conditions (1)-(12).

(60) TABLE-US-00009 TABLE 9 SD1 1.25 mm SD2 2.79 mm SD3 2.94 mm SD4 2.89 mm SD5 2.95 mm MT 3 mm L 2.5 mm ALD 12.82 mm f.sub.1/L1T −11.1065 SD5/TTL 0.284474 TTL/SD1 8.296 MT/L1T 3 MT/(SD2 + 0.259291 TTL/L 4.148 SD3 + SD4 + SD5) f/L 1.2 R.sub.11/L1T −11.2143 SD1/L1T 1.25 ALD/f 4.273333 f/L1T 3

(61) In addition, the lens assembly 3 of the third embodiment can meet the requirements of optical performance as seen in FIGS. 6A-6C. It can be seen from FIG. 6A that the field curvature of tangential direction and sagittal direction in the lens assembly 3 of the third embodiment ranges from −0.3 mm to 0.15 mm. It can be seen from FIG. 6B that the distortion in the lens assembly 3 of the third embodiment ranges from −1% to 4%. It can be seen from FIG. 6C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 3 of the third embodiment ranges from 0.17 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 3 of the third embodiment can be corrected effectively, and the resolution of the lens assembly 3 of the third embodiment can meet the requirement. Therefore, the lens assembly 3 of the third embodiment is capable of good optical performance.

(62) Referring to FIG. 7, the lens assembly 4 includes a cover glass CG4, a stop ST4, a first lens L41, a reflective element P4, a second lens L42, a third lens L43, a fourth lens L44, a fifth lens L45, and an optical filter OF4, all of which are arranged in order from an object side to an image side along an axis OA4. The reflective element P4 includes an incident surface S46, a reflective surface S47, and an exit surface S48, wherein the incident surface S46 and the exit surface S48 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S47 to change the propagation direction and imaged on an image plane IMA4. The image plane IMA4 and the exit surface S48 are parallel to each other. In the fourth embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(63) According to the foregoing, wherein: both of the object side surface S41 and image side surface S42 of the cover glass CG4 are plane surfaces; the first lens L41 is a biconcave lens, wherein the image side surface S45 is a concave surface, the object side surface S44 is an aspheric surface, and the image side surface S45 is a spherical surface; the second lens L42 is a meniscus lens, wherein the image side surface S410 is a concave surface; the third lens L43 is a biconvex lens, wherein the image side surface S412 is a convex surface; the fourth lens L44 is a meniscus lens, wherein the object side surface S413 is a convex surface and the image side surface S414 is a concave surface; the fifth lens L45 is a meniscus lens, wherein the object side surface S415 is a concave surface; and both of the object side surface S417 and image side surface S418 of the optical filter OF4 are plane surfaces.

(64) With the above design of the lenses, stop ST4, reflective element P4, and at least any one of the conditions (1)-(12) satisfied, the lens assembly 4 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(65) Table 10 shows the optical specification of the lens assembly 4 in FIG. 7.

(66) TABLE-US-00010 TABLE 10 Effective Focal Length = 3.02565 mm F-number = 2.48 Total Lens Length = 10.5536 mm Field of View = 64

degrees Radius of Effective Surface Curvature Thickness Focal Length Number (mm) (mm) Nd Vd (mm) Remark S41 ∞ 0.75 1.5168 64.167336 CG4 S42 ∞ 0.2 S43 ∞ 0.04 ST4 S44 -18.41896 1 2.003303 28.319976 -8.06672 L41 S45 15.00066 0.1 S46 ∞ 1.5 2.003303 28.319976 P4 Incident Surface S47 ∞ 1.5 2.003303 28.319976 P4 Reflective Surface S48 ∞ 0.05 P4 Exit Surface S49 6.270589 0.3865494 1.670993 19.243508 22.5545 L42 S410 10.36936 0.0329088 S411 6.004653 1.216863 1.5352 56.11 1.93727 L43 S412 -1.169108 0.035656 S413 2.130272 0.752001 1.661342 20.372904 -1.95201 L44 S414 0.694021 0.5252105 S415 -6.392578 0.8021414 1.5352 56.11 5.04751 L45 S416 -1.986759 0.906555 S417 ∞ 0.21 1.5168 64.167336 OF4 S418 ∞ 0.5457508

(67) The definition of aspheric surface sag z of each aspheric lens in table 10 is the same as that of in Table 1, and is not described here again.

(68) In the fourth embodiment, the conic constant k and the aspheric coefficients $A, B, C, D, E, F, G, H, I, J, K, L$ of each aspheric lens are shown in Table 11.

(69) TABLE-US-00011 TABLE 11 Surface A B C D E F Number k G H I J K L S44 -250.5742 -6.3450E-03 -6.1310E-03 -2.8135E-01 2.1338E+00 -6.5710E+00 9.3273E+00 -4.9263E+00 0 0 0 0 0 S49 9.779996 2.9625E-02 -1.4478E-01 4.9697E-02 2.1210E-02 -5.6629E-03 -1.2056E-02 5.5020E-03 3.5720E-05 -2.2394E-04 9.4550E-03 3.9876E-02 1.4460E-04 S410 6.996649 6.6154E-02 8.3783E-02 -4.6399E-02 -3.6966E-03 1.8047E-02 -7.4787E-04 -8.7826E-03 4.5230E-03 -6.8563E-04 4.1806E-02 -2.3781E-01 5.4746E-02 S411 -41.39938 -1.4690E-01 1.4950E-02 1.0397E-02 -6.2427E-02 6.2360E-03 1.6202E-02 -4.5689E-03 -1.1134E-03 4.2480E-04 6.1835E-02 -1.8995E-02 7.6550E-02 S412 -0.5172243 -4.2784E-01 6.4517E-02 -2.8731E-02 1.0820E-02 2.8240E-03 -3.3368E-03 -1.9001E-03 1.8360E-03 -3.3848E-04 6.1870E-02 2.0905E-01 -5.0095E-02 S413 -0.1859317 1.2874E-01 -1.1484E-01 4.7384E-02 1.0932E-02 -2.3375E-02 8.0380E-03 1.6350E-03 -1.4632E-03 2.2640E-04 3.4221E-02 -1.0974E-02 7.2770E-04 S414 -3.11992 3.2841E-01 1.7417E-01 -7.9922E-02 -5.1778E-02 5.0710E-02 9.3620E-03 -2.1357E-02 6.9430E-03 -6.2412E-04 -1.2247E-02 -5.0175E-01 9.8646E-02 S415 17.23926 -9.0582E-02 2.9264E-02 -3.1648E-02 -3.7874E-02 5.1248E-02 1.6746E-02 -3.6004E-02 1.4723E-02 -1.9936E-03 2.7770E-02 -1.5815E-02 -3.6162E-02 S416 0 -1.9250E-01 -6.6542E-02 1.1020E-01 -5.8663E-02 1.5858E-02 -2.2353E-04 1.0899E-02 -1.0281E-02 2.4550E-03 2.7378E-02 1.1183E-01 -9.0166E-02

(70) Table 12 shows the parameters and condition values for conditions (1)-(12) in accordance with the fourth embodiment of the invention. It can be seen from Table 12 that the lens assembly **4** of the fourth embodiment satisfies the conditions (1)-(12).

(71) TABLE-US-00012 TABLE 12 SD1 1.519168 mm SD2 2.831159 mm SD3 2.913121 mm SD4 2.925538 mm SD5 2.642572 mm MT 3 mm L 2.6 mm ALD 12.83156 mm f.sub.1/L1T -8.06672 SD5/TTL 0.250395 TTL/SD1 6.94696 MT/L1T 3 MT/(SD2 + 0.265196 TTL/L 4.059077 SD3 + SD4 + SD5) f/L 1.163712 R.sub.11/L1T -18.419 SD1/L1T 1.519168 ALD/f 4.240926 f/L1T 3.02565

(72) In addition, the lens assembly **4** of the fourth embodiment can meet the requirements of optical performance as seen in FIGS. **8A-8C**. It can be seen from FIG. **8A** that the field curvature of tangential direction and sagittal direction in the lens assembly **4** of the fourth embodiment ranges from -0.2 mm to 0.08 mm. It can be seen from FIG. **8B** that the distortion in the lens assembly **4** of the fourth embodiment ranges from -0.1% to 1.8%. It can be seen from FIG. **8C** that the modulation transfer function of tangential direction and sagittal direction in the lens assembly **4** of the fourth embodiment ranges from 0.01 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly **4** of the fourth embodiment can be corrected effectively, and the resolution of the lens assembly **4** of the fourth embodiment can meet the requirement. Therefore, the lens assembly **4** of the fourth embodiment is capable of good optical performance.

(73) Referring to Table 13, Table 14, Table 16, Table 17, Table 19, Table 20, Table 22, and Table 23, wherein Table 13, Table 16, Table 19, and Table 22 show optical specification in accordance with a fifth, sixth, seventh, and eighth embodiments of the invention respectively and Table 14, Table 17, Table 20, and Table 23 show aspheric coefficients of each aspheric lens in Table 13, Table 16, Table 19, and Table 22 respectively. FIG. **9**, FIG. **11**, FIG. **13**, and FIG. **15** are lens layout and optical path diagrams of the lens assemblies in accordance with the fifth, sixth, seventh, and eighth embodiments of the invention respectively.

(74) The first lenses **L51**, **L61**, **L71**, **L81** are meniscus lenses with positive refractive power and made of glass material, wherein the object side surfaces **S52**, **S62**, **S72**, **S82** are concave surfaces, the image side surfaces **S53**, **S63**, **S73**, **S83** are convex surfaces, and both of the object side surfaces **S52**, **S62**, **S72**, **S82** and image side surfaces **S53**, **S63**, **S73**, **S83** are aspheric surfaces. The object side surfaces **S52**, **S62**, **S72**, **S82** are concave at the near optical axis which helps to achieve the design specification requirements such as miniaturization and shortening total lens length.

(75) The reflective elements **P5**, **P6**, **P7**, **P8** are made of glass or plastic material, wherein the incident surfaces **S54**, **S64**, **S74**, **S84** are plane surfaces, the reflective surfaces **S55**, **S65**, **S75**, **S85** are plane surfaces, and the exit surfaces **S56**, **S66**, **S76**, **S86** are plane surfaces. The reflective element can also be a prism or a reflective mirror. The reflective element can also only include one reflective surface when the reflective element is a reflective mirror. With the configuration of the reflective element, the total lens length of the lens assembly can be prevented from being too long, which causes excessive occupation of the limited space of the lens assembly and results to poor space utilization, thus reducing the volume of the lens assembly and increasing the space usage.

(76) The second lenses **L52**, **L62**, **L72**, **L82** are meniscus lenses with negative refractive power and made of plastic material, wherein the object side surfaces **S57**, **S67**, **S77**, **S87** are convex surfaces, the image side surfaces **S58**, **S68**, **S78**, **S88** are concave surfaces, and both of the object side surfaces **S57**, **S67**, **S77**, **S87** and image side surfaces **S58**, **S68**, **S78**, **S88** are aspheric surfaces. The object side surfaces **S57**, **S67**, **S77**, **S87** are convex at the near optical axis which makes the light of each field of view entering the lens assembly uniformly and corrects the off-axis image bending phenomenon.

(77) The third lenses **L53**, **L63**, **L73**, **L83** are with refractive power and made of plastic material, wherein both of the object side surfaces **S59**, **S69**, **S79**, **S89** and image side surfaces **S510**, **S610**, **S710**, **S810** are aspheric surfaces.

(78) The fourth lenses **L54**, **L64**, **L74**, **L84** are with positive refractive power and made of plastic material, wherein the image side surfaces **S512**, **S612**, **S712**, **S812** are convex surfaces and both of the object side surfaces **S511**, **S611**, **S711**, **S811** and image side surfaces **S512**, **S612**, **S712**, **S812** are aspheric surfaces.

(79) The fifth lenses **L55**, **L65**, **L75**, **L85** are meniscus lenses with refractive power and made of plastic material, wherein the object side surfaces **S513**, **S613**, **S713**, **S813** are convex surfaces, the image side surfaces **S514**, **S614**, **S714**, **S814** are concave surfaces, and both of the object side surfaces **S513**, **S613**, **S713**, **S813** and image side surfaces **S514**, **S614**, **S714**, **S814** are aspheric surfaces.

(80) In addition, the lens assemblies **5**, **6**, **7**, **8** satisfy at least any one of the conditions (1)-(5) and the following conditions:

$$3 < (f_{\text{sub.1}} + f) / \text{SD1} < 5; \quad (13)$$

$$-6 < R_{\text{sub.11}} / R_{\text{sub.52}} < -0.5; \quad (14)$$

$$1 < f_{\text{sub.1}} / L1T < 4; \quad (15)$$

$$1 \text{ mm} \cdot \text{sup.2} < f_{\text{sub.1}} \times f_{\text{sub.4}} < 11 \text{ mm} \cdot \text{sup.2}; \quad (16)$$

$$-12 \text{ mm} \cdot \text{sup.2} < f_{\text{sub.2}} \times f_{\text{sub.4}} < -1 \text{ mm} \cdot \text{sup.2}; \quad (17)$$

$$0.5 < f_{\text{sub.1}} / L < 2.5; \quad (18)$$

(81) wherein $f_{\text{sub.1}}$ is an effective focal length of the first lenses **L51**, **L61**, **L71**, **L81** for the fifth to eighth embodiments, $f_{\text{sub.2}}$ is an effective focal length of the second lenses **L52**, **L62**, **L72**, **L82** for the fifth to eighth embodiments, $f_{\text{sub.4}}$ is an effective focal length of the fourth lenses **L54**, **L64**, **L74**, **L84** for the fifth to eighth embodiments, f is an effective focal length of the lens assemblies **5**, **6**, **7**, **8** for the fifth to eighth embodiments, SD1 is an effective optical diameter of the first lenses **L51**, **L61**, **L71**, **L81** for the fifth to eighth embodiments, $R_{\text{sub.11}}$ is a radius of curvature of the object side surfaces **S52**, **S62**, **S72**, **S82** of the first lenses **L51**, **L61**, **L71**, **L81** for the fifth to eighth embodiments, $R_{\text{sub.52}}$ is a radius of curvature of the image side surfaces **S514**, **S614**, **S714**, **S814** of the fifth lenses **L55**, **L65**, **L75**, **L85** for the fifth to eighth embodiments, $L1T$ is a thickness along the axes **OA5**, **OA6**, **OA7**, **OA8** of the first lenses **L51**, **L61**, **L71**, **L81** for the fifth to eighth embodiments, and L is an interval from the object side

L64 S612 -1.75577254 0.06010465 S613 0.91392142 0.34308493 1.535218308 56.11525368 3.941 L65 S614 1.39777675 0.5 S615 ∞ 0.21 1.52 64.2 OF6 S616 ∞ 0.32787752

(99) The definition of aspheric surface sag z of each aspheric lens in table 16 is the same as that of in Table 1, and is not described here again.

(100) In the sixth embodiment, the conic constant k and the aspheric coefficients $A, B, C, D, E, F, G, H, I$ of each aspheric lens are shown in Table 17.

(101) TABLE-US-00017 TABLE 17 Surface A B C Number k G H I D E F S62 4.432740462 0.010764626 0.21968427 -1.885123398 5.929518046 38.42501078 -259.9227724 422.9781425 0 0 S63 0.926620236 0.09605128 0.191269033 -0.53102089 1.163868838 -0.063169413 -1.947773676 1.766822598 0 0 S67 58.47570306 0.009977518 0.062854517 0.176525119 -0.68184053 1.051334074 -0.776719634 0.222747751 0 0 S68 0.361834987 0.12279405 -0.010838413 0.016986132 0.01830016 0.00378226 -0.016344568 0.006310691 0 0 S69 -28.59459118 0.01713649 -0.120013871 0.026837936 0.036648217 -0.011110027 -0.010743742 0.004555882 0 0 S610 -37.08756609 0.20380597 -0.134444622 0.070263813 -0.013100146 -0.01677969 -0.0031976 0.004611458 0 0 S611 57.33320315 -0.082667797 0.029695028 -0.011337166 0.008089738 -0.000245386 -0.005151514 0.001617653 0 0 S612 0.063401978 -0.074214845 -0.055768944 0.009649235 0.007764448 -0.002074531 -0.000614818 0.000296749 0 0 S613 -2.776066849 0.000788809 0.137363316 0.009312183 0.003889615 -0.010694935 0.001128218 -3.323E-05 0 0 S614 -0.654390403 0.154564126 0.070186096 -0.007295362 -0.001770642 5.7785E-07 -0.001745787 0.000414664 0 0

(102) Table 18 shows the parameters and condition values for conditions (1)-(5) and (13)-(18) in accordance with the sixth embodiment of the invention. It can be seen from Table 18 that the lens assembly 6 of the sixth embodiment satisfies the conditions (1)-(5) and (13)-(18).

(103) TABLE-US-00018 TABLE 18 SD1 1.42 mm SD2 2.38 mm SD3 2.72 mm SD4 2.78 mm SD5 3 mm MT 2.11 mm L 2.11 mm (f.sub.1 + f)/SD1 3.49 R.sub.11/R.sub.52 -1.25 f.sub.1/L1T 2.57 R.sub.11/L1T -1.72 MT/(SD2 + 0.19 TTL/SD1 4.52 SD3 + SD4 + SD5) MT/L1T 2.09 f.sub.1 $\times$ f.sub.4 9.95 mm.sup.2 f.sub.2 $\times$ f.sub.4 -11.12 mm.sup.2 f.sub.1/L 1.23

(104) In addition, the lens assembly 6 of the sixth embodiment can meet the requirements of optical performance as seen in FIGS. 12A-12C. It can be seen from FIG. 12A that the field curvature of tangential direction and sagittal direction in the lens assembly 6 of the sixth embodiment ranges from -0.1 mm to 0.4 mm. It can be seen from FIG. 12B that the distortion in the lens assembly 6 of the sixth embodiment ranges from -1% to 3%. It can be seen from FIG. 12C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 6 of the sixth embodiment ranges from 0.29 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 6 of the sixth embodiment can be corrected effectively, and the resolution of the lens assembly 6 of the sixth embodiment can meet the requirement. Therefore, the lens assembly 6 of the sixth embodiment is capable of good optical performance.

(105) Referring to FIG. 13, the lens assembly 7 includes a stop ST7, a first lens L71, a reflective element P7, a second lens L72, a third lens L73, a fourth lens L74, a fifth lens L75, and an optical filter OF7, all of which are arranged in order from an object side to an image side along an axis OA7. The incident surface S74 and the exit surface S76 of the reflective element P7 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S75 to change the propagation direction and imaged on an image plane IMA7. The image plane IMA7 and the exit surface S76 are parallel to each other. In the seventh embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(106) According to the foregoing, wherein: the third lens L73 is a biconvex lens with positive refractive power, wherein the object side surface S79 is a convex surface and the image side surface S710 is a convex surface; the fourth lens L74 is a meniscus lens, wherein the object side surface S711 is a concave surface; the fifth lens L75 is with positive refractive power; and both of the object side surface S715 and image side surface S716 of the optical filter OF7 are plane surfaces.

(107) With the above design of the lenses, stop ST7, reflective element P7, and at least any one of the conditions (1)-(5) and (13)-(18) satisfied, the lens assembly 7 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(108) Table 19 shows the optical specification of the lens assembly 7 in FIG. 13.

(109) TABLE-US-00019 TABLE 19 Effective Focal Length = 1.661 mm F-number = 2.4 Total Lens Length = 6.484 mm Field of View = 75 degrees Radius of Effective Surface Curvature Thickness Focal Length Number (mm) (mm) Nd Vd (mm) Remark S71 ∞ 0.1 ST7 S72 -1.76E+00 1.108 1.85 40.4 2.87 L71 S73 -1.33E+00 0.040 S74 ∞ 1.000 2.00 28.3 P7 Incident Surface S75 ∞ 1.000 2.00 28.3 P7 Reflective Surface S76 ∞ 0.029 P7 Exit Surface S77 -4.158 0.252 1.67 19.2 -2.2 L72 S78 -1.073 0.073 S79 -2.780 0.782 1.53 56.1 4.96 L73 S710 57.898 0.045 S711 3.525 0.735 1.53 56.1 2.07 L74 S712 0.909 0.020 S713 -1.176 0.556 1.53 56.1 14.92 L75 S714 -1.150 0.534 S715 ∞ 0.21 1.52 64.2 OF7 S716 ∞ 0.10

(110) The definition of aspheric surface sag z of each aspheric lens in table 19 is the same as that of in Table 1, and is not described here again.

(111) In the seventh embodiment, the conic constant k and the aspheric coefficients $A, B, C, D, E, F, G, H, I$ of each aspheric lens are shown in Table 20.

(112) TABLE-US-00020 TABLE 20 Surface A B C Number k G H I D E F S72 1.58E+01 3.90E-01 -7.54E-01 3.54E+01 -8.55E+01 -1.57E+03 3.05E+03 8.05E+04 1.77E+05 -5.77E+06 S73 2.50E+00 1.68E-01 5.77E-01 -1.02E-01 -1.29E+01 4.15E+01 4.35E+02 -3.23E+03 7.92E+03 -1.36E+04 S77 1.06E+00 4.98E-01 -4.89E-01 5.00E-01 -6.70E+00 2.41E+01 -4.01E+01 3.59E+01 -1.68E+01 6.53E+00 S78 1.20E+00 6.54E-01 -7.19E-01 3.04E-01 1.54E-01 -9.50E-02 -9.93E-02 7.29E-02 -4.38E-03 -4.34E-03 S79 1.21E+00 -3.89E-01 5.64E-01 -3.05E-01 -8.70E-02 4.62E-02 1.72E-01 -1.45E-01 2.15E-02 1.46E-02 S710 1.25E+00 5.93E-02 1.57E-02 1.14E-02 -1.38E-03 -5.13E-03 -2.13E-03 -9.30E-04 3.05E-04 5.90E-04 S711 1.28E+00 -2.95E-02 -9.33E-03 -2.86E-03 -1.48E-03 1.01E-03 2.72E-04 -2.26E-04 -4.39E-05 5.55E-04 S712 1.32E+00 -2.12E-01 -4.22E-02 -1.17E-01 1.06E-01 3.39E-02 -4.84E-02 -2.37E-02 3.21E-02 -1.55E-02 S713 1.35E+00 8.31E-02 -5.96E-02 1.31E-03 1.36E-01 -1.44E-01 6.50E-02 7.13E-03 -1.73E-02 8.12E-03 S714 1.40E+00 5.96E-01 -3.16E-01 7.74E-02 4.64E-02 -1.47E-02 -9.43E-03 2.21E-03 1.91E-03 -1.32E-03

(113) Table 21 shows the parameters and condition values for conditions (1)-(5) and (13)-(18) in accordance with the seventh embodiment of the invention. It can be seen from Table 21 that the lens assembly 7 of the seventh embodiment satisfies the conditions (1)-(5) and (13)-(18).

(114) TABLE-US-00021 TABLE 21 SD1 1.18 mm SD2 2.4 mm SD3 2.5 mm SD4 2.64 mm SD5 2.8 mm MT 2 mm L 2.148 mm (f.sub.1 + f)/SD1 3.84 R.sub.11/R.sub.52 -1.53 f.sub.1/L1T 2.59 R.sub.11/L1T -1.59 MT/(SD2 + 0.19 TTL/SD1 5.49 SD3 + SD4 + SD5) MT/L1T 1.81 f.sub.1 $\times$ f.sub.4 5.94 mm.sup.2 f.sub.2 $\times$ f.sub.4 -4.56 mm.sup.2 f.sub.1/L 1.34

(115) In addition, the lens assembly 7 of the seventh embodiment can meet the requirements of optical performance as seen in FIGS. 14A-14C. It can be seen from FIG. 14A that the field curvature of tangential direction and sagittal direction in the lens assembly 7 of the seventh embodiment ranges from -0.14 mm to 0.05 mm. It can be seen from FIG. 14B that the distortion in the lens assembly 7 of the seventh embodiment ranges from 0% to 4%. It can be seen from FIG. 14C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 7 of the seventh embodiment ranges from 0.40 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 7 of the seventh embodiment can be corrected effectively, and the resolution of the lens assembly 7 of the seventh embodiment can meet the requirement. Therefore, the lens assembly 7 of the seventh embodiment is capable of good optical performance.

(116) Referring to FIG. 15, the lens assembly 8 includes a stop ST8, a first lens L81, a reflective element P8, a second lens L82, a third lens L83, a fourth lens L84, a fifth lens L85, and an optical filter OF8, all of which are arranged in order from an object side to an image side along an axis OA8. The incident surface S84 and the exit surface S86 of the reflective element P8 are perpendicular to each other. In operation, the light from the object side is reflected by the reflective surface S85 to change the propagation direction and imaged on an image plane IMA8. The image plane IMA8 and the exit surface S86 are parallel to each other. In the eighth embodiment, the reflective element takes a prism as an example but is not limited thereto. For example, the reflective element may be a reflective mirror which only includes a reflective surface.

(117) According to the foregoing, wherein: the third lens L83 is a meniscus lens with negative refractive power and the object side surface S89 is a concave surface and the image side surface S810 is a convex surface; the fourth lens L84 is a biconvex lens, wherein the object side surface S811 is a convex surface; the fifth lens L85 is with negative refractive power; and both of the object side surface S815 and image side surface S816 of the optical filter OF8 are plane surfaces.

(118) With the above design of the lenses, stop ST8, reflective element P8, and at least any one of the conditions (1)-(5) and (13)-(18) satisfied, the lens assembly 8 can have an effective shortened total lens length, an effective decreased outer diameter, an effective increased resolution, an effective corrected aberration, an effective corrected chromatic aberration, and is capable of an easy manufacture process.

(119) Table 22 shows the optical specification of the lens assembly 8 in FIG. 15.

(120) TABLE-US-00022

TABLE 22	Effective Focal Length = 2.36 mm	F-number = 2.4	Total Lens Length = 5.653 mm	Field of View = 75 degrees
Radius of Effective Surface	Curvature Thickness	Focal Length	Number (mm)	(mm)
Nd	Vd (mm)	Remark	S81 ∞ 0.1	ST8 S82 -2.56 1.000 1.72 53.5
2.28	L81 S83 -1.17 0.040	S84 ∞ 1.000 2.00 28.3	P8 Incident Surface	S85 ∞ 1.000 2.00 28.3
P8 Reflective Surface	S86 ∞ 0.020	P8 Exit Surface	S87 -22.284 0.220 1.67 19.2 -2.57	L82 S88 -1.611 0.138
S89 6.865 0.504 1.53 56.1 -24.3	L83 S810 14.843 0.020	S811 -3.742 0.546 1.53 56.1 1.54	L84 S812 1.011 0.020	S813 -0.843 0.220 1.53 56.1 -3.15
L85 S814 -0.511 0.615	S815 ∞ 0.21 1.52 64.2	OF8 S816 ∞ 0.10		

(121) The definition of aspheric surface sag z of each aspheric lens in table 22 is the same as that of in Table 1, and is not described here again.

(122) In the eighth embodiment, the conic constant k and the aspheric coefficients A, B, C, D, E, F, G, H, I of each aspheric lens are shown in Table 23.

(123) TABLE-US-00023

TABLE 23	Surface A	B	C	Number k	G	H	I	D	E	F	S82	2.38E+01	2.00E-01	-3.55E+00	3.52E+01	1.13E+01	-1.36E+03									
-7.08E+02	7.70E+04	-3.66E+05	1.08E+06	S83	1.20E+00	1.51E-01	3.61E-01	-2.63E-02	-1.68E+01	5.32E+01	4.66E+02	-3.30E+03	7.51E+03	-1.19E+04	S87	1.03E+00	2.10E-01	-3.09E-01	7.84E-01	-6.93E+00	2.39E+01	-4.01E+01	3.61E+01	-1.67E+01	6.33E+00	S88
1.09E+00	3.67E-01	-5.46E-01	2.90E-01	1.64E-01	-9.81E-02	-1.14E-01	5.34E-02	-1.03E-02	9.09E-03	S89	1.11E+00	-3.94E-01	4.03E-01	-2.50E-01	-5.07E-02	4.52E-02	1.61E-01	-1.57E-01	1.25E-02	8.34E-03	S810	1.12E+00	-9.10E-02	1.62E-01	1.14E-01	-4.90E-02
-5.43E-02	-2.04E-02	-4.77E-03	4.80E-03	1.12E-02	S811	1.15E+00	-9.78E-03	9.50E-02	-5.83E-02	7.34E-02	4.54E-02	-2.35E-02	-4.95E-02	-1.83E-02	5.75E-02	S812	1.27E+00	-3.23E-01	6.66E-02	-1.20E-01	8.85E-02	2.86E-02	-4.87E-02	-2.27E-02	3.23E-02	
-1.51E-02	S813	1.34E+00	5.91E-01	-2.34E-01	-1.24E-01	1.80E-01	-1.22E-01	6.03E-02	1.88E-03	-1.91E-02	1.15E-02	S814	1.49E+00	6.22E-01	-4.01E-01	7.20E-02	4.47E-02	-1.01E-02	-7.93E-03	1.13E-03	1.26E-03	-6.40E-04				

(124) Table 24 shows the parameters and condition values for conditions (1)-(5) and (13)-(18) in accordance with the eighth embodiment of the invention. It can be seen from Table 24 that the lens assembly 8 of the eighth embodiment satisfies the conditions (1)-(5) and (13)-(18).

(125) TABLE-US-00024

TABLE 24	SD1 1.25 mm	SD2 2.19 mm	SD3 2.23 mm	SD4 2.55 mm	SD5 2.98 mm	MT 2 mm	L 2.04 mm	(f.sub.1 + f)/SD1	3.71 R.sub.11/R.sub.52	-5.00 f.sub.1/L1T	2.28 R.sub.11/L1T	-2.56 MT/(SD2 + 0.20 TTL/SD1	4.52 SD3 + SD4 + SD5)	MT/L1T	2 f.sub.1 $\times$ f.sub.4	3.51 mm.sup.2	f.sub.2 $\times$ f.sub.4	-3.96 mm.sup.2	f.sub.1/L	1.12

(126) In addition, the lens assembly 8 of the eighth embodiment can meet the requirements of optical performance as seen in FIGS. 16A-16C. It can be seen from FIG. 16A that the field curvature of tangential direction and sagittal direction in the lens assembly 8 of the eighth embodiment ranges from -0.06 mm to 0.10 mm. It can be seen from FIG. 16B that the distortion in the lens assembly 8 of the eighth embodiment ranges from 0% to 3%. It can be seen from FIG. 16C that the modulation transfer function of tangential direction and sagittal direction in the lens assembly 8 of the eighth embodiment ranges from 0.34 to 1.0. It is obvious that the field curvature and the distortion of the lens assembly 8 of the eighth embodiment can be corrected effectively, and the resolution of the lens assembly 8 of the eighth embodiment can meet the requirement. Therefore, the lens assembly 8 of the eighth embodiment is capable of good optical performance.

(127) While the invention has been described by way of example and in terms of the preferred embodiment(s), it is to be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

Claims

1. A lens assembly consisting of: a cover glass; a stop; a first lens which is with refractive power and comprises a concave surface facing an object side along an axis; a second lens which is with refractive power and comprises a convex surface facing the object side along the axis; a third lens which is with refractive power; a fourth lens which is with refractive power and comprises a convex surface facing the object side along the axis and a concave surface facing an image side along the axis; a fifth lens which is with refractive power; an optical filter; and a reflective element which comprises a reflective surface; wherein the cover glass, the stop, the first lens, the second lens, the third lens, the fourth lens, the fifth lens, and the optical filter are arranged in order from the object side to the image side along the axis; wherein the reflective element is disposed between the first lens and the fifth lens; wherein the lens assembly satisfies the following conditions:
 $0.5 < SD1/L1T < 3$; wherein SD1 is an effective optical diameter of the first lens and L1T is a thickness along the axis of the first lens.
2. The lens assembly as claimed in claim 1, wherein: the first lens is with negative refractive power; the second lens is with positive refractive power; the third lens is with positive refractive power; the fourth lens is with negative refractive power; and the fifth lens is with positive refractive power.
3. The lens assembly as claimed in claim 1, wherein: the first lens further comprises a plane surface or a concave surface facing the image side along the axis; the third lens comprises a convex surface facing the object side along the axis; and the fourth lens is a meniscus lens.
4. The lens assembly as claimed in claim 1, wherein the fifth lens comprises a convex surface facing the image side along the axis and another convex surface or a concave surface facing the object side along the axis.
5. The lens assembly as claimed in claim 1, wherein: the second lens is a meniscus lens and further comprises a concave surface facing the image side along the axis; and the third lens is a biconvex lens and further comprises a convex surface facing the image side along the axis.
6. The lens assembly as claimed in claim 1, wherein: the reflective element further comprises an incident surface facing the object side along the axis and an exit surface facing the image side along the axis; and the lens assembly satisfies at least any one of the following conditions:
 $0.1 < SD5/TTL < 0.6$;
 $2 \text{ mm} < L < 6 \text{ mm}$;
 $0.5 < MT/L1T < 10$;
 $0 < MT/(SD2+SD3+SD4+SD5) < 1$;
 $0 < f/L1T < 5$;
 $1 < TTL/L < 5$;
 $0 < f/L < 2$;
 $-20 < R_{sub.11}/L1T < 0$;
 $2 < ALD/f < 8$;
 $-12 < f_{sub.1}/L1T < 0$; wherein MT is an interval from the incident surface to the exit surface along the axis, L1T is the thickness along the axis of the first lens, SD2 is an effective optical diameter of the second lens, SD3 is an effective optical diameter of the third lens, SD4 is an effective optical diameter of the fourth lens, and SD5 is an effective optical diameter of the fifth lens, L is an interval from the object side surface of the first lens to the reflective surface along the axis, TTL is the interval from the object side surface of the first lens to the image plane along the axis, f is an effective

focal length of the lens assembly, $R_{sub.11}$ is a radius of curvature of the object side surface of the first lens, ALD is a total sum of the effective optical diameter of all lenses of the lens assembly, and $f_{sub.1}$ is an effective focal length of the first lens.

7. The lens assembly as claimed in claim 1, wherein the lens assembly satisfies the following condition:

$$3 < (f_{sub.1} + f) / SD1 < 5;$$

$$-6 < R_{sub.11} / R_{sub.52} < -0.5;$$

$$1 < f_{sub.1} / L1T < 4;$$

$$1 \text{ mm} < f_{sub.2} < f_{sub.1} \times f_{sub.4} < 11 \text{ mm};$$

$$-12 \text{ mm} < f_{sub.2} \times f_{sub.4} < -1 \text{ mm};$$

$0.5 < f_{sub.1} / L < 2.5$; wherein $R_{sub.11}$ is a radius of curvature of the object side surface of the first lens, $R_{sub.52}$ is a radius of curvature of an image side surface of the fifth lens, f is an effective focal length of the lens assembly, $f_{sub.2}$ is an effective focal length of the second lens, $f_{sub.4}$ is an effective focal length of the fourth lens, $L1T$ is the thickness along the axis of the first lens, $SD1$ is the effective optical diameter of the first lens, $f_{sub.1}$ is an effective focal length of the first lens and L is the interval from the object side surface of the first lens to the reflective surface along the axis.

8. A lens assembly consisting of: a cover glass; a stop; a first lens which is with refractive power and comprises a concave surface facing an object side along an axis; a second lens which is a meniscus lens with refractive power and comprises a convex surface facing the object side along the axis and a concave surface facing the image side along the axis; a third lens which is a biconvex lens and comprises a convex surface facing the image side along the axis; a fourth lens which is with refractive power; a fifth lens which is with refractive power; an optical filter; and a reflective element which comprises a reflective surface; wherein the cover glass, the stop, the first lens, the second lens, the third lens, the fourth lens, the fifth lens, and the optical filter are arranged in order from the object side to an image side along the axis; wherein the reflective element is disposed between the first lens and the fifth lens; wherein the lens assembly satisfies the following conditions:

$3.5 < TTL / SD1 \leq 14$; wherein $SD1$ is an effective optical diameter of the first lens, and TTL is an interval from an object side surface of the first lens to an image plane along the axis.

9. The lens assembly as claimed in claim 8, wherein: the first lens is with positive refractive power; the second lens is with negative refractive power; the fourth lens is with positive refractive power.

10. The lens assembly as claimed in claim 8, wherein: the reflective element further comprises an incident surface facing the object side along the axis and an exit surface facing the image side along the axis; and the lens assembly satisfies at least any one of the following conditions:

$$0.1 < SD5 / TTL < 0.6;$$

$$2 \text{ mm} < L < 6 \text{ mm};$$

$$0.5 < MT / L1T < 10;$$

$$0 < MT / (SD2 + SD3 + SD4 + SD5) < 1;$$

$$0 < f / L1T < 5;$$

$$1 < TTL / L < 5;$$

$$0 < f / L < 2;$$

$$-20 < R_{sub.11} / L1T < 0;$$

$$2 < ALD / f < 8;$$

$$-12 < f_{sub.1} / L1T < 0;$$

$0.5 < SD1 / L1T < 3$; wherein L is an interval from the object side surface of the first lens to the reflective surface along the axis, $SD5$ is an effective optical diameter of the fifth lens, TTL is the interval from the object side surface of the first lens to the image plane along the axis, MT is an interval from the incident surface to the exit surface along the axis, $L1T$ is the thickness along the axis of the first lens, $SD1$ is an effective optical diameter of the first lens, $SD2$ is an effective optical diameter of the second lens, $SD3$ is an effective optical diameter of the third lens, $SD4$ is an effective optical diameter of the fourth lens, f is an effective focal length of the lens assembly, $R_{sub.11}$ is a radius of curvature of the object side surface of the first lens, ALD is a total sum of the effective optical diameter of all lenses of the lens assembly, and $f_{sub.1}$ is an effective focal length of the first lens.

11. The lens assembly as claimed in claim 8, wherein: the first lens further comprises a convex surface facing the image side along the axis.

12. The lens assembly as claimed in claim 8, wherein: the third lens comprises a convex surface facing the object side along the axis; and the fifth lens is with positive refractive power.

13. The lens assembly as claimed in claim 8, wherein: the lens assembly satisfies at least any one of the following conditions:

$$0.1 < SD5 / TTL < 0.6;$$

$$2 \text{ mm} < L < 6 \text{ mm};$$

$$0 < f / L1T < 5;$$

$$1 < TTL / L < 5;$$

$$0 < f / L < 2;$$

$$-20 < R_{sub.11} / L1T < 0;$$

$$2 < ALD / f < 8;$$

$$-12 < f_{sub.1} / L1T < 0;$$

$$-20 < R_{sub.11} / L1T < 0;$$

$$3 < (f_{sub.1} + f) / SD1 < 5;$$

$$-6 < R_{sub.11} / R_{sub.52} < -0.5;$$

$$1 < f_{sub.1} / L1T < 4;$$

$$1 \text{ mm} < f_{sub.2} < f_{sub.1} \times f_{sub.4} < 11 \text{ mm};$$

$$-12 \text{ mm} < f_{sub.2} \times f_{sub.4} < -1 \text{ mm};$$

$0.5 < f_{sub.1} / L < 2.5$; wherein $R_{sub.11}$ is a radius of curvature of the object side surface of the first lens, $L1T$ is a thickness along the axis of the first lens, TTL is the interval from the object side surface of the first lens to the image plane along the axis, $SD1$ is the effective optical diameter of the first lens, L is an interval from the object side surface of the first lens to the reflective surface along the axis, $SD5$ is an effective optical diameter of the fifth lens, f is an effective focal length of the lens assembly, ALD is a total sum of the effective optical diameter of all lenses of the lens assembly, $f_{sub.1}$ is an effective focal length of the first lens, $R_{sub.52}$ is a radius of curvature of an image side surface of the fifth lens, $f_{sub.2}$ is an effective focal length of the second lens, and $f_{sub.4}$ is an effective focal length of the fourth lens; the reflective element further comprises an incident surface facing the object side along the axis and an exit surface facing the image side along the axis; and the lens assembly satisfies at least any one of the following conditions:

$$0.5 < MT / L1T < 10;$$

$0 < MT / (SD2 + SD3 + SD4 + SD5) < 1$; wherein MT is an interval from the incident surface to the exit surface along the axis, $L1T$ is the thickness along the axis of the first lens, $SD2$ is an effective optical diameter of the second lens, $SD3$ is an effective optical diameter of the third lens, $SD4$ is an effective optical diameter of the fourth lens, and $SD5$ is the effective optical diameter of the fifth lens.

14. The lens assembly as claimed in claim 8, wherein the lens assembly satisfies at least any one of the following conditions:

$$3 < (f_{sub.1} + f) / SD1 < 5;$$

$$-6 < R_{sub.11} / R_{sub.52} < -0.5;$$

$1 < f_{\text{sub.1}}/L1T < 4$;

$1 \text{ mm.sup.2} < f_{\text{sub.1}} \times f_{\text{sub.4}} < 11 \text{ mm.sup.2}$;

$-12 \text{ mm.sup.2} < f_{\text{sub.2}} \times f_{\text{sub.4}} < -1 \text{ mm.sup.2}$;

$0.5 < f_{\text{sub.1}}/L < 2.5$; wherein $R_{\text{sub.11}}$ is a radius of curvature of the object side surface of the first lens, $R_{\text{sub.52}}$ is a radius of curvature of an image side surface of the fifth lens, f is an effective focal length of the lens assembly, $f_{\text{sub.1}}$ is an effective focal length of the first lens, $f_{\text{sub.2}}$ is an effective focal length of the second lens, $f_{\text{sub.4}}$ is an effective focal length of the fourth lens, $L1T$ is a thickness along the axis of the first lens, L is an interval from the object side surface of the first lens to the reflective surface along the axis, and $SD1$ is the effective optical diameter of the first lens.

15. The lens assembly as claimed in claim 8, wherein the fourth lens comprises a convex surface facing the image side along the axis and another convex surface or a concave surface facing the object side along the axis.

16. The lens assembly as claimed in claim 8, wherein the fifth lens is a meniscus lens and comprises a convex surface facing the object side along the axis and a concave surface facing the image side along the axis.
